

TOPICS

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0.1 Zou-He BCs

Given the velocity $\vec{u} = (u, v)$, the *macroscopic velocity* now splits into two components

$$\begin{aligned}\rho u &= \sum_i c f_i e_{ix}, \\ &= f_1 - f_3 + f_5 - f_6 - f_7 + f_8, \\ f_1 + f_5 + f_8 &= \rho u + (f_3 + f_6 + f_7); \end{aligned}$$

$$\begin{aligned}\rho v &= \sum_i c f_i e_{iy}, \\ &= f_2 - f_4 + f_5 + f_6 - f_7 - f_8, \\ f_5 - f_8 &= \rho v - f_2 + f_4 - f_6 + f_7. \end{aligned}$$

With

$$f_1 - f_1^{\text{eq}} = f_3 - f_3^{\text{eq}} \implies f_1 = f_3 + (f_1^{\text{eq}} - f_3^{\text{eq}}),$$

the D2Q9 model now becomes

$$f_i^{\text{eq}} = w_i \rho \left[1 + 3(\vec{e}_i \cdot \vec{u}) + \frac{9}{2}(\vec{e}_i \cdot \vec{u})^2 - \frac{3}{2}\vec{u}^2 \right].$$

Since $\vec{e}_1 = (1, 0)$, $\vec{e}_3 = (-1, 0)$, and $w_1 = w_3 = \frac{1}{9}$, we obtain

$$\begin{aligned}f_1^{\text{eq}} - f_3^{\text{eq}} &= \frac{1}{9} \rho [3u - (-3u)], \\ &= \frac{2}{3} \rho u, \\ \implies f_1 &= f_3 + \frac{2}{3} \rho u. \end{aligned}$$

Substitute $f_1 = f_3 + \frac{2}{3} \rho u$, we have

$$f_5 + f_8 = f_6 + f_7 + \frac{2}{3} \rho u.$$

Thus, we can get f_5 with

$$\begin{aligned}f_5 &= \frac{1}{2} [(f_5 + f_8) + (f_5 - f_8)], \\ &= f_7 - \frac{1}{2}(f_2 - f_4) + \frac{1}{6} \rho u + \frac{1}{6} \rho v. \end{aligned}$$

Similarly, f_8 is

$$\begin{aligned} f_8 &= \frac{1}{2} [(f_5 + f_8) - (f_5 - f_8)], \\ &= f_6 + \frac{1}{2}(f_2 - f_4) + \frac{1}{6}\rho u - \frac{1}{6}\rho v. \end{aligned}$$

1 Steady Plane Poiseuille Flow

The incompressible Navier-Stokes equations are

$$\begin{aligned} \nabla \cdot \vec{u} &= 0; \\ \rho \left(\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} \right) &= -\nabla p + \mu \nabla^2 \vec{u}. \end{aligned}$$

We assume the steady plane Poiseuille flow only in x -direction, and varies only with y . That is,

$$\vec{u} = (u(y), 0).$$

Thus,

$$\begin{aligned} \frac{\partial u}{\partial t} &= 0; \\ u \frac{\partial u}{\partial x} &= 0; \\ v \frac{\partial u}{\partial y} &= 0. \end{aligned}$$

This is due to the steady flow, u has no x -dependence, and $v = 0$. The x -momentum equation becomes

$$0 = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} \implies \mu \frac{\partial^2 u}{\partial y^2} = \frac{\partial p}{\partial x},$$

where the pressure gradient is

$$\frac{\partial p}{\partial x} = \frac{P_1 - P_0}{L}.$$

Solve for the differential equation, we obtain

$$\frac{d^2 u}{dy^2} = \frac{1}{\mu} \frac{P_1 - P_0}{L}.$$

After one integration, we have

$$\frac{du}{dy} = \frac{P_1 - P_0}{\mu L} y + C_1.$$

After another integration, we get

$$u(y) = \frac{P_1 - P_0}{2\mu L} y^2 + C_1 y + C_2.$$

Apply the boundary conditions, the coefficients are

$$\begin{aligned} u(0) = 0 &\implies C_2 = 0; \\ u(H) = 0 &\implies \frac{P_1 - P_0}{2\mu L} H^2 + C_1 H = 0; \\ &\implies C_1 = -\frac{P_1 - P_0}{2\mu L} H. \end{aligned}$$

As a result, the solution is

$$u(y) = \frac{\Delta p}{2\mu L} y(y - H).$$

The criterion for steady state is given by

$$\frac{\sum_{ij} |u_{ij}^{n+1} - u_{ij}^n|}{\sum_{ij} |u_{ij}^{n+1}|} \leq 5.0 \times 10^{-9}.$$

The flow field will be no longer evolving in time if the relative change between the two consecutive iterations becomes extremely small.